

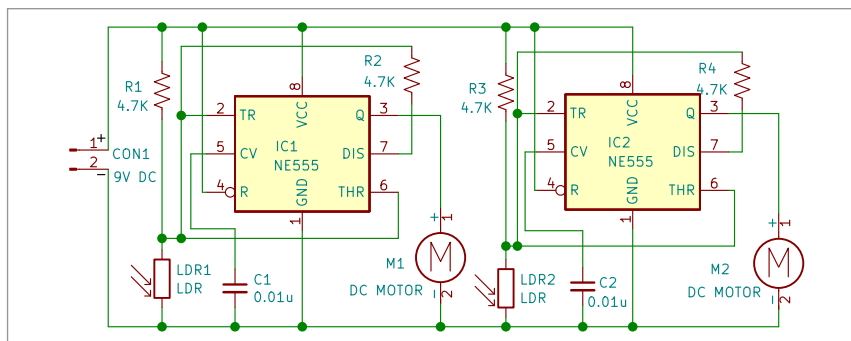


Fig. 3: Circuit diagram



Fig. 4: Robot ready to use

nected to the 555 timer IC. When it's dark, the LDR's resistance increases, and the voltage at the voltage divider drops below 0.8V, causing the 555 timer IC output to go low to turn off the motor. When there is enough light, the voltage at the discharge pin 7 goes above 0.8V,

and the 555 IC output goes high to turn on the motor.

Construction and testing

Before assembling the circuit based on the circuit diagram shown in Fig. 3, also refer to Fig. 2. This will give an exact idea about the construction of the robot. Assemble the circuit directly on the breadboard and use a cabinet and wheels in such a

way that the robot can smoothly move when light falls on the LDRs. LDRs are mounted on the top of the breadboard in such a way that light should fall on them for the proper movement of the robot.

After proper assembly, your light-activated robot is ready to use. When light falls on LDR1 or LDR2, respective motors rotate accordingly, resulting in the robot's movement. An image of a ready-to-use robot on a breadboard is shown in Fig. 4. **EFY**

PARTS LIST

Semiconductors:

IC1, IC2 - 555 timer

Resistors (all 1/4-watt, $\pm 5\%$ carbon):

R1-R4 - 4.7-kilo-ohm

Capacitors:

C1, C2 - 0.01 μ F polyester

Miscellaneous:

LDR1, LDR2 - Light-dependent resistor

M1, M2 - 12V DC motor

- Robot assembly

- Two wheels

- One small moving wheel

- Jumper wires, male-to-male

- Breadboard

- 9V DC battery/adaptor

(L to R) Manmohan Singh, Vivek Gowda, Ramiksha Shetty, and Sahana P from Electronics and Communication Department, Sapthagiri College of Engineering, Bengaluru



Anyone who stops learning is old, whether at twenty or eighty.

—Henry Ford

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